CropSense

Diagnosis Report

Date: 2026-03-08
Location: 2021 S First St, Champaign, IL 61820

CONFIDENCE: HIGH

Blue-Green Ear Mold (Penicillium Ear Rot)

Penicillium spp.

Crop: corn (Zea mays) | fungal | Severity: Moderate

9.3C	91.0%	18.8km/h	7.6C	0.3%	100.0%
Temp	Humidity	Wind	Soil T.	Soil M.	Cloud

Recent Precipitation: 0.0 mm

Weather is Not Favorable for Spread

Pressure: Unknown

What AI Observed:

Ear tip shows dense bluegreen fuzzy mold with a few adjacent kernels discolored and slightly shriveled; remainder of ear appears mostly sound.

ear | kernels | silk region | tip of the ear

Action Plan

Do These First:

1. Prioritize harvesting affected fields or patches if grain is at or near maturity; segregate visibly moldy ears during harvest.
2. Avoid handling/harvesting when ears are wet with dew or fog; schedule operations for late morning/afternoon once surfaces are dry to limit spore spread.
3. Use PPE (gloves, N95/respirator) when handling moldy ears to avoid inhalation of spores and potential mycotoxins.

Treatment Options:

Timely harvest

High if implemented quickly before further humid periods.

Harvest promptly to limit further mold growth and spore dispersal from the ear tip; adjust combine to reduce kernel damage and remove fines.

Timing: With cool, humid weather, do not delay. If forecast calls for additional humidity/rain in 2448 hours, advance harvest schedule.

Rapid drying and proper storage

High for preventing further mold development postharvest.

Dry grain immediately after harvest to 15% moisture for short-term storage and 1314% for longer-term; clean bins, remove fines, and aerate to keep grain cool and dry.

Timing: Because ambient air is cool and very humid, natural-air drying will be slow; use heated air/adequate airflow and begin drying immediately after harvest.

Grain segregation and testing

High value for safety and marketability.

Keep suspect loads separate; test for mycotoxins before feeding or marketing if mold is evident.

Timing: As soon as grain is harvested; do not delay testing if moldy grain will be fed.

What Confirmed This Diagnosis:

- Distinct bluegreen fuzzy mold concentrated at the ear tip
- Localized shriveling/necrosis of kernels beneath mold
- Growth on and around silk remnants and exposed tip
- Association with cool, humid conditions and damaged/exposed kernels

Prevention & Evidence

Preventive Measures:

- Select hybrids with good tip cover and tight husks; consider traits and proven tolerance to ear rots in local trials.
- Manage insects (e.g., ear-feeding pests) and birds that injure kernels and facilitate mold entry.
- Promote uniform pollination and plant health to reduce exposed tip kernels; avoid lodging and harvest delays.
- Rotate crops and manage residue to reduce inoculum; maintain good field drainage.
- Thoroughly clean and sanitize storage structures; monitor grain temperature and moisture routinely.

Scout & Monitor

Under continued cool, humid conditions, scout multiple field locations (20 ears per area) focusing on ear tips and silk attachment points; estimate percent ears affected and severity. After harvest, check bins daily for crusting, heating, or musty odors; track grain moisture and temperature and aerate proactively when ambient air is sufficiently dry.

Could Also Be...

Aspergillus ear rot (Aspergillus flavus)
Likelihood: Less likely: typically yellowgreen/olive powdery mold; favored by warmer conditions.

Fusarium ear rot (Fusarium verticillioides/F. proliferatum)
Likelihood: Possible but less consistent: usually whitish to pink mold on scattered kernels rather than dense bluegreen tip mold.

Cladosporium ear rot
Likelihood: Possible but less consistent: typically olivegreen to black sooty mold rather than bluegreen.

Sources:

[1] **Ear and Kernel Rots of Corn (BP-59-W)**
Purdue Extension (2018)
https://www.extension.purdue.edu/extmedia/BP/BP-59-W.pdf

[2] **Common Corn Ear Rots (MF3010)**
Kansas State Research and Extension (2016)
https://bookstore.ksre.ksu.edu/pubs/MF3010.pdf

[3] **Corn Ear Rot Diseases**
Iowa State University Extension and Outreach (2014)
https://crops.extension.iastate.edu/cropnews/2014/09/corn-ear-rot-diseases

Visual symptoms strongly indicate Penicillium ear rot, but other ear molds can co-occur. Laboratory confirmation is recommended to verify pathogen(s) and assess any mycotoxin risks before feeding or marketing.